### 7.5.3.1 Polariser and analyser

Let us consider an unpolarised beam of light. The vibrations can be in all possible directions perpendicular to the direction of propagation as shown in Figure 7.27. When this light passes through a polaroid $P_1$ the vibrations are restricted to only one plane. The emergent beam can be further passed through another polaroid $P_2$. If the polaroid $P_2$ is rotated by keeping the ray of light as axis, for a particular position of $P_2$ the intensity is maximum. When the polaroid $P_2$ is rotated further, the intensity starts decreasing. There is complete extinction of the light when $P_2$ is rotated through 90°. On further rotating $P_2$, the light reappears and the intensity increases and becomes maximum at 90°. The light coming out from polaroid $P_1$ is said to be plane polarised. **The Polaroid (here $P_1$) which polarises the light passing through it is called a *polariser*. The polaroid (here $P_2$) which is used to examine whether a light is polarised or not is called an *analyser*.**

If the intensity of the unpolarised light is $I$ then the intensity of polarised light will be $\left(\frac{I}{2}\right)$. The other half of intensity is restricted by the polariser.

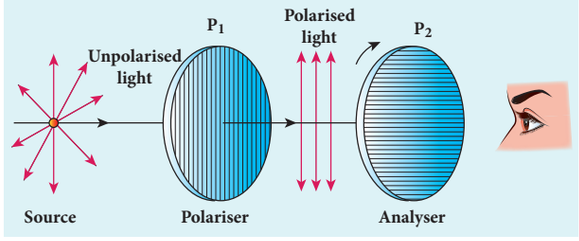



**Figure 7.27** Polariser and analyser

### 7.5.3.2 Plane and partially polarised light

**A light is said to be *plane polarised* if the intensity varies from maximum to zero for every 90° rotation of the analyser** as shown in the graph in Figure 7.28(a). This is because the vibrations are allowed in one direction and completely restricted in the perpendicular direction. On the other hand, **if the intensity of light varies between maximum and minimum (not zero) for every 90° rotation of the analyser, the light is said to be *partially polarised* light** as shown in the graph in Figure 7.28(b). This is because the light is not fully restricted in that particular direction which remains as a minimum intensity.

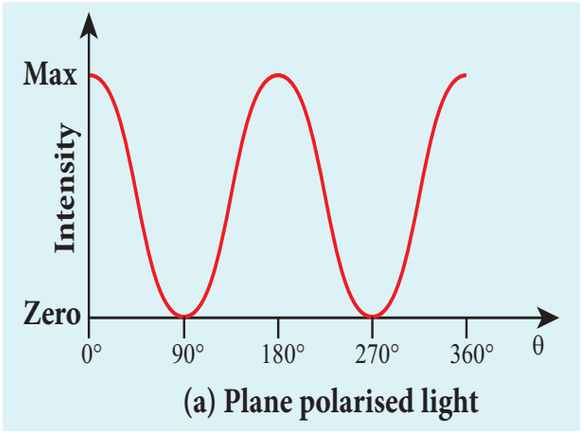



**Figure 7.28** Intensity variation in (a) plane and (b) partially polarised light

### 7.5.3.3 Malus' law

In 1809, French Physicist E.N Malus discovered that when a beam of plane polarised light of intensity $I_0$ is incident



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